01 / PROBLEM AND CONTRIBUTION

2.3 V LDO regulator

EEE 433/591 · Analog IC Design | Fall 2025 | Individual design

RESULT AT A GLANCE

Held the simulated DC output within an 8 mV span across four load conditions by designing an OTA-driven PMOS regulator and evaluating its feedback response.

The engineering question

How does a CMOS regulator maintain an approximately 2.3 V output as the load changes from 0.1 to 50 mA?

Situation

The load on a regulator changes while its output must stay close to the required voltage. This project evaluates a 2.3 V LDO from a 2.5 V supply across a 0.1-50 mA load range.

Task

Build the OTA-controlled feedback loop and distinguish steady-state regulation from the response to a sudden load change.

My contribution

LDO circuit design and DC, AC, and transient evaluation for the analog IC class project.

Tools and concepts

Analog IC design / LDO / Feedback / Cadence Virtuoso / Transient analysis

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

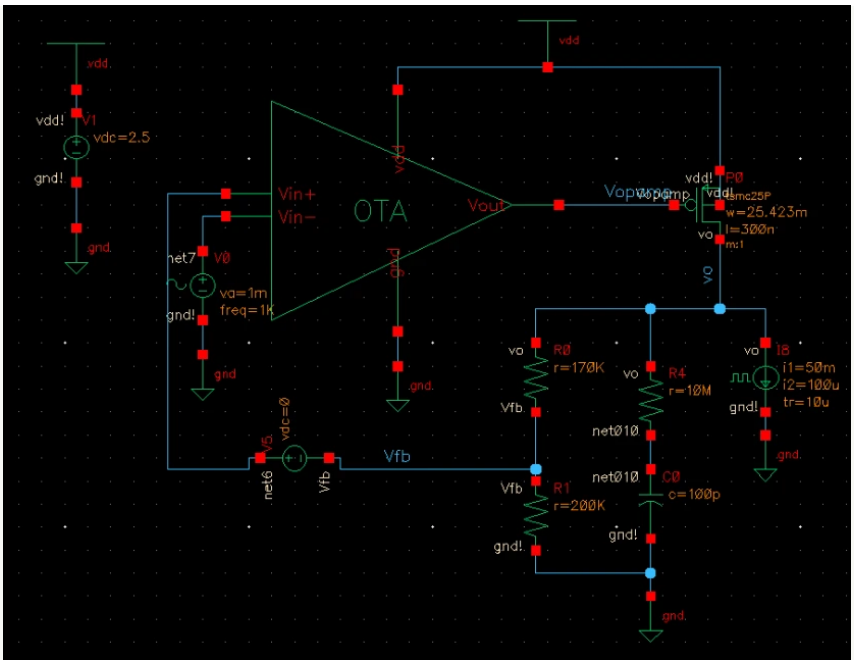
Closed the feedback loop around an OTA error amplifier, a PMOS pass transistor, and a resistive divider. The OTA adjusts the pass-device gate to bring the feedback voltage toward the reference.

02 / Implementation step

Compared four DC load conditions, then applied a load step to examine droop, overshoot, and recovery. Operating-point checks relate the pass-device current and terminal voltages to the changing load.

Closing the control loop

The feedback divider senses the output; the OTA compares that signal with the reference and controls the PMOS gate. The DC sweep examines the equilibrium output, whereas the step experiment examines the time-dependent return toward equilibrium.



Closed-loop LDO schematic

OTA, PMOS pass device, and feedback network.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

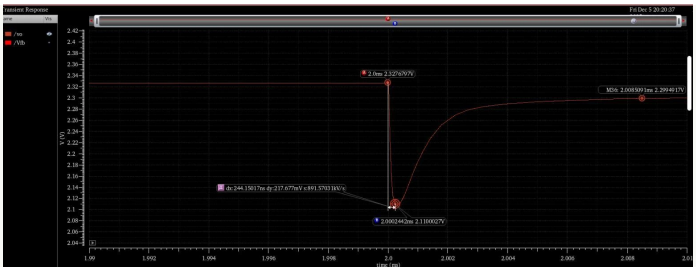
Supply voltage	2.5 V
DC load range	0.1-50 mA
DC output span	8 mV

Reported result

At a 2.5 V supply, the reported DC output spans 2.315-2.323 V across the four load conditions: an 8 mV range. The project brief includes the closed-loop schematic, load sweep, and transient waveform.

Two distinct evaluations

The 2.315-2.323 V output range is the difference between reported steady-state readings. It cannot be used as a transient droop or overshoot figure. The waveform should be read together with the applied load transition and its time scale.



Load-step transient response

Simulation response to the reported 0.1 to 50 mA load step.

Scope and limitations

The 8 mV figure describes static output variation across the DC sweep. The load-step waveform provides a separate view of transient droop, overshoot, and recovery.

Possible next steps - proposed, not completed results

- Reconcile AC and settling-time summaries before publishing numeric loop-stability or settling specifications.
- Record transient droop, overshoot, and recovery with explicit load-step and output-capacitance conditions.

Source material: EEE 433/591 final report, pp. 8-16; approved LDO brief.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.

Project files and documentation: Low-Dropout-Regulator-LDO-